Importance of early successional habitat to ruffed grouse and American woodcock

by Daniel R. Dessecker and Daniel G. McAuley

Abstract Ruffed grouse (*Bonasa umbellus*) and American woodcock (*Scolopax minor*) provide millions of days of recreation each year for people in the eastern United States (U.S). These popular game birds depend on early successional forest habitats throughout much of the year. Ruffed grouse and woodcock populations are declining in the eastern United States as an abundance of shrub-dominated and young forest habitats decrease in most of the region. Continued decreases in early successional forest habitats are likely on nonindustrial private forest lands as ownership fragmentation increases and tract size decreases and on public forest lands due to societal attitudes toward proactive forest management, especially even-age treatments.

Key Words American woodcock, aspen, *Bonasa umbellus,* early successional habitat, even-age management, ruffed grouse, *Scolopax minor*

Ruffed grouse (*Bonasa umbellus*) and American woodcock (*Scolopax minor*) depend on shrub-dominated and young forest habitats. The high stem densities characteristic of these habitats protect them from predators and enable local populations to attain levels substantially greater than on landscapes dominated by mature forest (Sepik and Dwyer 1982, Gullion 1984*a*). Historically, these early successional habitats were established through periodic disturbance. Fires of aboriginal and "natural" origins were the primary disturbance agents in much of the eastern United States, although insect infestation and wind also affected vegetation structure and composition (Little 1974). Intensity, extent, and frequency of fires varied spatially due to landscape conditions (slope, aspect, etc.). In addition, fire frequency varied temporally in response to changes in climate and changes in Native American population density and distribution (Dessecker 1997, Hamel and Buckner 1998).

The ruffed grouse is the most popular upland game bird throughout much of its range in the eastern United States. Where ruffed grouse populations are cyclic, hunter numbers commonly rise and fall with local populations. During the most recent cyclic high in the late 1990s, approximately 120,000 hunters spent 1,000,000 days afield in each of Michigan, Minnesota, and Wisconsin (Berg 2000, Dhuey 2000, Whitcomb et al. 2000). Total annual ruffed grouse harvest likely approaches 1,000,000 birds in each of these states during the peak of the 10-year cycle. Approximately 300,000 ruffed grouse are harvested annually during cyclic lows.

Ruffed grouse population, hunter effort, and harvest data are scarce outside of the Great Lakes region.

Address for Daniel R. Dessecker: Ruffed Grouse Society, P.O. Box 2, Rice Lake, WI 54868, USA; e-mail: rgsdess@chibardun.net. Address for Daniel G. McAuley: United States Geological Survey, Patuxent Wildlife Research Center, 5768 South Annex A, Orono, ME 04469-5768, USA.

Gullion (unpublished data) surveyed state resource agencies across the United States and estimated that Maine, New York, and Pennsylvania each harvested approximately 450,000 ruffed grouse in the mid-1980s. Indices of ruffed grouse hunter effort and harvest suggest that both are declining throughout the eastern United States. These data are consistent with survey results documenting a 36% decline in numbers of small-game hunters in the United States between 1985 and 1996 (United States Department of Interior [USDI] 1997).

The American woodcock also is a popular game bird throughout eastern North America. In the 1980s, woodcock provided approximately 3.4 million days of recreational hunting annually (USDI 1990). At that time, U.S. hunters harvested an estimated 1.1 million woodcock annually (Straw et al. 1994), making woodcock among the top 10 species of migratory game birds harvested in the Atlantic and Mississippi flyways. Current estimates of hunter effort and harvest of American woodcock are available from the recently established Harvest Information Program (HIP), although estimates from this survey are imprecise. These data suggest that in 1998–99, 128,000 hunters spent 574,000 days afield and harvested 435,000 woodcock (United States Fish and Wildlife Service [USFWS], Office of Migratory Bird Management [MBMO], unpublished data). States leading in hunter effort and harvest include Louisiana, Michigan, Minnesota, Wisconsin, New York, and Maine.

In this paper we 1) review the current status, recent trends, and factors affecting ruffed grouse and American woodcock populations in the eastern United States; 2) summarize use of early successional habitats by these 2 species; 3) present habitat management recommendations; and 4) discuss the outlook for future trends of these species.

Distribution and status

Ruffed grouse

The ruffed grouse is North America's most widely distributed gallinaceous bird (Johnsgard 1973). Ruffed grouse are found throughout much of the eastern United States, yet are common only where extensive tracts of forest dominate the landscape. Ruffed grouse are common in the northern Great Lakes region and where suitable habitats exist in the Northeast and the central and southern Appalachian Mountains. The southern extreme of the ruffed grouse range coincides with the southern edge of the Appalachians in northeast Georgia. Ruffed grouse are generally rare below 460 m elevation in the southeast portion of their range, although habitats that appear suitable exist in the Piedmont from Louisiana east to Georgia and north through Virginia.

There is no range-wide population survey of ruffed grouse, but some states monitor populations or harvest rates. Male ruffed grouse drum in the spring to attract females. Drumming-male surveys count all males heard in the early morning along 10- or 15-stop routes and can provide an index of local populations (Gullion 1966). Drumming-male densities typically reach 1 to 2 birds/40 ha in the central hardwood forests of the Midwest, the central and southern Appalachians, and in northern hardwood forests in the northern tier of states (Thompson and Dessecker 1997). The aspen (*Populus* spp.) forests of the Great Lakes region can support 4–8 drumming males/40 ha (Kubisiak 1985).

Early successional habitats are by nature ephemeral.

We obtained survey data on trends in drumming males from 5 states: Minnesota (Berg 2000), Wisconsin (Dhuey 2000), Indiana (Backs 2000), Ohio (R. Stoll, Ohio Department of Natural Resources, unpublished data), and Kentucky (J. Sole, Kentucky Department of Fish and Wildlife, unpublished data). These data suggest long-term declines for 3 of these 5 states, though only 2 correlations between grouse abundance and year are significant (Minnesota, r=–0.22, P=0.227; Wisconsin, r=0.04, P=0.824; Indiana, r=–0.89, P<0.001; Ohio, r=–0.727, P<0.001; Kentucky, r=0.61, P=0.143). Hunter flush-rate data (birds flushed/unit of hunter effort) are collected from cooperating hunters in some states. We obtained flush-rate data from Ohio, North Carolina (T. Sharpe, North Carolina Wildlife Resources Commission, unpublished data), Tennessee (M. Gudlin, Tennessee Wildlife Resources Agency, unpublished data), Virginia (Norman 2000), and West Virginia (W. Lesser, West Virginia Department of Natural Resources, unpublished data). Correlations between grouse abundance and year indicate long-term declines for 2 of these 5 states (Ohio, r=–0.454, P=0.03; North Carolina, r=0.297, P=0.38; Tennessee, r=–0.75, P<0.01; Virginia, r=–0.096, P=0.67; West Virginia, r=0.009, P=0.96). The cyclic nature of ruffed grouse populations in northern latitudes (Figure 1) has been well documented (Keith 1963) and complicates interpretation of trends for these populations.

Woodcock

The American woodcock is a member of the shorebird family that inhabits forested and mixed forest–urban–agricultural areas from Manitoba east to Newfoundland and Labrador and south to Florida, the Gulf of Mexico,

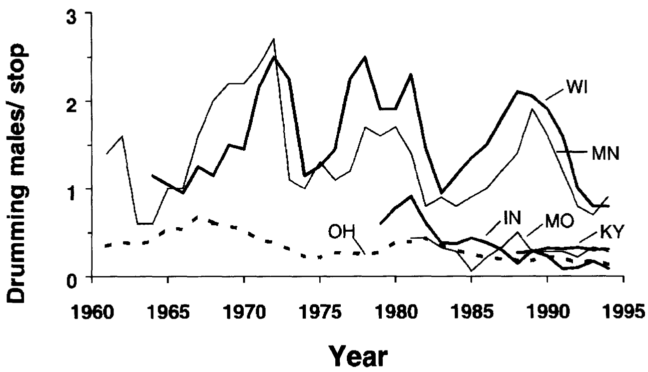


Figure 1. Trends and relative abundance of ruffed grouse based on drumming males heard/stop along multiple-stop survey routes.

and eastern Texas. The northern limit of breeding is indistinct, but may extend to James Bay or southern Hudson Bay (Sheldon 1967, Keppie and Whiting 1994, Straw et al. 1994). In its northern range, the woodcock is one of the earliest ground-nesting species. Woodcock breed most commonly north of 40° N latitude, but there are records of nests in every state and province within its range (Keppie and Whiting 1994). In the south, nesting begins in February, whereas in the north nesting begins in April and May and extends into June or early July (Pettingill 1936; United States Geological Survey, unpublished data).

Woodcock are migratory and winter in southern states where snow cover or ground frost is rare. In the East, Cape May, New Jersey, and Cape Charles, Virginia, are major staging areas for woodcock during migration, especially in fall. Most birds arrive on wintering areas by mid-December (Keppie and Whiting 1994). Areas of high woodcock concentrations recorded during the National Audubon Society's Christmas Bird Count are eastern Texas to central Louisiana, the coastal plain of South Carolina, and the lower Delmarva peninsula to eastern Virginia (Straw et al. 1994).

Woodcock are managed as 2 regional populations, the Eastern and the Central Management Units (Owen et al. 1977). This delineation is justified by band recovery data indicating little crossover of birds between the regions (Martin et al. 1969, Krohn et al. 1974). Reliable indices of population size, productivity, harvest, and distribution of woodcock are difficult to obtain (Bruggink and Kendall 1995). Because of their small size, cryptic coloration, and preference for dense vegetation, it is impractical to census woodcock populations. However, woodcock populations are monitored with the North American Singing-ground Survey (SGS) and the Wing-collection Survey (WCS). In 1968, about 1,500 routes were located randomly across the entire northern breeding range of the woodcock (see Sauer and Bortner [1991] for a review of implementation and analysis of the SGS). The SGS takes advantage of the male woodcock's conspicuous courtship display and provides an index to number of displaying male woodcock present in the population. The surveys count all calling males heard along a 10-stop route that is run after sunset. The short-term (1990–2000) and long-term (1968–2000) trends in both management units are declining (Eastern Unit r^2 =0.926, P< 0.01; Central Unit r^2 =0.903, P<0.01 [Kelley 2000]; Figure 2). In the Eastern Management Unit, number of males heard along routes declined at about 2.3%/year since 1968 and by 3.5%/year over the last 10 years. Trends in the Central Management Unit declined 1.6%/year since 1968 and 3.1%/year since 1990.

The WCS is conducted every year in the U.S. and Canada to provide data on reproductive success of woodcock, information on the chronology and distribution of the harvest, and data on hunting success (Kelley 2000). Age and sex can be determined from wing-plumage characteristics (Martin 1964) and the ratio of immature birds/adult female in the survey sample is an index of recruitment. In the U.S. in 1999, recruitment indices in the Eastern (1.1) and the Central (1.2) units were the least on record and were >25% below the 1998 index and >29% below the long-term regional averages of about 1.7 (Kelley 2000). In Canada, the 10-year (1988–97) average recruitment index was 3.1 in Nova Scotia, 2.4 in New

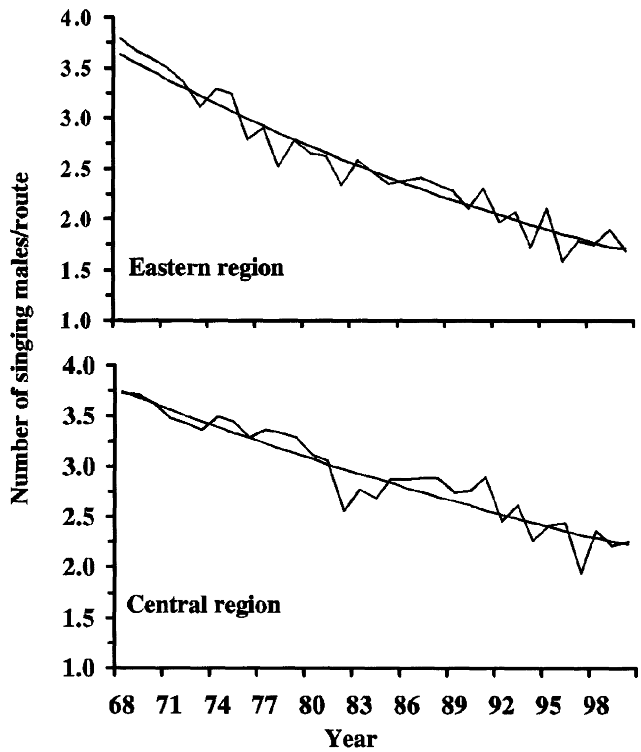


Figure 2. Long-term trends and annual indices of the number of male woodcock heard on the Singing-ground Survey, 1968–2000.

Brunswick, and 1.4 in Ontario. In Canada, the proportion of young in the survey has been declining over the long term (1982–97) in Ontario (P=0.06), whereas changes in Nova Scotia (-) and New Brunswick (+) were not significant (Bateman 1999).

Woodcock harvest is declining in the U.S. and Canada. The estimated harvest in Canada of 45,558 in 1997 was 57% below the 10-year mean (Bateman 1999). In the U.S., estimates of total harvest are imprecise (see Straw et al. 1994), but results of surveys indicate that woodcock harvest and number of woodcock hunters have declined since the early 1980s (Straw et al. 1994, Kelley 2000). Owen et al. (1977) estimated the U.S. harvest at 1.5 million birds; Straw et al. (1994) estimated the harvest in 1991 at 1.1 million, whereas the recent estimates from the HIP survey indicated harvest in 1999 was 435,000 woodcock (USFWS MBMO, unpublished data).

Cause of population declines

Habitat loss and degradation are the predominant factors affecting ruffed grouse and woodcock population trends. Studies suggest that ground-nesting songbirds may currently be experiencing low reproductive rates, possibly due to landscape-level changes in habitat conditions (Robinson et al. 1995). Ruffed grouse and woodcock are ground-nesting birds. Ongoing ruffed grouse research in 7 central Appalachian states has documented relatively high nest success (69%), yet very low chick survival (25%) to age 5 weeks (Appalachian Cooperative Grouse Research Project, unpublished data). Declines in young forest habitats and the isolation of these habitats in some landscapes may be limiting ruffed grouse and woodcock recruitment and therefore population densities.

Habitat use

Ruffed grouse

Ruffed grouse are early successional forest specialists. Optimum habitats for ruffed grouse include young (6- to 15-year-old), even-age deciduous stands that typically support 20–25,000 woody stems/ha (Gullion 1984*a*; Kubisiak 1985; Stoll et al. 1999; Dimmick et al.1998). These habitats are available to grouse for approximately 1 decade because stem densities decrease rapidly through natural thinning as succession proceeds.

Although commonly identified as an "edge" species, ruffed grouse association with habitat edges largely reflects their use of various interspersed forest habitats at different times of the year and their use of marginal edge habitats where quality habitat is lacking. Ruffed grouse typically avoid hard-contrast edges. Gullion (1984*b*:73) states: "...extensive use of forest edges by ruffed grouse provides the best indication of how unsatisfactory a forest habitat has become for these birds."

Ruffed grouse use young stands of many different deciduous forest types throughout North America. However, aspen forests can support ruffed grouse population densities that greatly exceed those attained in other forest communities (Thompson and Dessecker 1997). Regeneration stem densities in recently clearcut aspen stands commonly reach levels that provide excellent protective cover for ruffed grouse. In addition, the dormant flower bud from mature male aspen trees is an important source of food for grouse in winter and early spring.

Young deciduous forest and shrub-dominated old-field habitats protect ruffed grouse from predators throughout the year. Dense stands are especially important to drumming males in spring (Stoll et al. 1979, Boyd 1990).

Habitats used for nesting appear to be variable; nesting hens can be found in a wide variety of habitats, although commonly in forest habitats that are older and more open than those frequented during other times of the year. Ruffed grouse broods are seldom found far from dense cover. Quality brood habitat includes small forest openings with a substantial shrub component. These openings can provide an important source of insects for developing chicks during their first 4–6 weeks of life (Hollifield and Dimmick 1995). Brood habitat is often relatively mesic, typically on north or east slopes in hilly terrain, or in riparian areas (Godfrey 1975, Kubisiak 1978, Thompson et al. 1987).

Ruffed grouse use a wide variety of available foods throughout the year. However, winter food availability and quality may be limiting factors for ruffed grouse populations in the central and southern Appalachians (Servello and Kirkpatrick 1987). In these regions,

American woodcock depend on early successional habitats and moist soils.

The ruffed grouse is the most popular game bird throughout much of the eastern United States.

succulent herbaceous vegetation, typically found on relatively mesic sites, is an important component of ruffed grouse diets.

Woodcock

Habitats used by woodcock vary with activity, time of day, and season, but like ruffed grouse they are an early successional species. They are not restricted to specific plant assemblages (Keppie and Whiting 1994) as long as the habitat provides the necessary early successional structure (Straw et al. 1994). Dense young forest or shrub-dominated habitats on moist soils are ideal (Keppie and Whiting 1994). Moist soils are an important component of quality woodcock habitat as they ensure that earthworms, which comprise nearly 80% of their diet (Sperry 1940), are at or near the soil surface and available to foraging woodcock.

In spring, males need openings called singing grounds to perform courtship displays and attract females for mating. Vegetative composition of singing grounds varies locally and throughout the range and does not likely determine use (Dwyer et al. 1988, Sepik et al. 1993). More likely the quality of the adjacent habitat for nesting and brood rearing determines singing-ground use by males. At night during summer, many birds use clearings, such as blueberry (*Vaccinium* spp.) barrens, pastures, recently harvested woodlands, and plantations, for roosting (Dunford and Owen 1973, Sepik et al. 1981, Sepik and Derleth 1993). Many of these same fields are used for singing grounds in spring.

Woodcock nest in a variety of habitats. Nests and broods are found in young to mixed-age forests, although they prefer young hardwood stands (Mendall and Aldous 1943), especially aspen (Gregg and Hale 1977, McAuley et al. 1996). Typically, numbers of trees >7.6 cm diameter (400–783/ha) and basal area (6–9.5 m^2/ha) are low, whereas density of saplings <7.6 cm (1,400–4,500/ha) and shrubs (13, 500–49,250/ha) is high (Bourgeois 1977, Coon et al. 1982, Parris 1986, Kinsley and Storm 1988, McAuley et al. 1996).

During summer, young hardwoods or older stands with a dense understory, particularly alder (*Alnus* spp.), provide daytime cover for feeding (Morgenweck 1977, Rabe 1977, Hudgins et al. 1985). In northern breeding areas, conifer stands are used rarely, except during drought when they may be critical for survival (Sepik et al. 1983). Diurnal habitats in fall and on migration are young hardwood stands on moist soils with dense shrubs (Keppie and Whiting 1994).

In winter, a variety of habitats are used diurnally, especially bottomland hardwoods, upland mixed pine–hardwoods, and recently burned stands of longleaf pine (*Pinus palustris*), although young hardwood stands are preferred (Glasgow 1958, Britt 1971, Dyer and Hamilton 1977, Krementz and Pendleton 1994). In pine stands, preferred microhabitats include depressions or drainages dominated by deciduous species. Mature stands of bottomland hardwoods are used if the canopy is sufficiently open to allow an understory of saplings, vines, and forbs to develop (Roberts et al. 1984). In Louisiana, Dyer and Hamilton (1977) reported that bottomland hardwoods used by woodcock contained dense stands of small-diameter trees. Kroll and Whiting (1977) in eastern Texas found substantial woodcock use of 2-year-old clearcut pine stands and pole-sized mixed pine–hardwood stands that had a high basal area of pine and abundant deciduous shrubs. Causey (1989) reported high variability in habitats used by woodcock, yet most stands had a well-developed shrub understory.

Habitat management

Ruffed grouse

Early successional habitats are by nature ephemeral. On landscapes where it is impractical to allow the return of natural fires or introduce prescribed fires, commercial timber harvests and other proactive habitat management practices must be implemented at regular intervals (approximately every 10 years) to ensure a continuous supply of quality ruffed grouse habitat on the landscape. Forest types that reach biological or economic maturity more rapidly can be managed using shorter rotations, thereby increasing amount of ruffed grouse habitat that is available on the landscape at any one time. Reductions in the proportion of a management unit where forest management is practiced can reduce ruffed grouse habitat potential.

Even-age silvicultural systems (clearcut, seed tree, shelterwood) are the most appropriate methods to create ruffed grouse habitat. These methods remove sufficient canopy from the parent stand to result in enough understory development to provide protective cover for ruffed grouse. Group-selection treatments can produce stem densities comparable to clearcut regeneration harvests (Weigel and Parker 1995), but patch sizes are generally too small to provide secure cover for ruffed grouse. Selection methods may be beneficial in riparian areas or other areas where understory development is desired yet even-age management is precluded.

Regeneration harvests that retain low levels of residual basal area are called clearcut with reserves, modified shelterwood harvests, or deferment cuts. In general, the greatest amount of overstory removal will yield the greatest degree of understory development. Retention of a limited number of residual trees may not affect regeneration stem densities in developing stands. Smith et al. (1989) found similar stem densities 5 years post-treatment in clearcut central hardwood stands and stands with <4.9 m^2/ha of residual basal area. Aspen is extremely shade-intolerant. Perala (1977) showed that as little as 2.5–3.7 m^2/ha of residual basal area can reduce aspen regeneration growth by 40%. Residual basal areas of 2.5 m^2/ha can reduce aspen regeneration stem densities after the first growing season by 29% (D. M. Stone , United States Forest Service [USFS], unpublished data).

The diameter distribution of residual trees also can significantly affect regeneration stem densities. For example, retaining approximately 16 38-cm-diameter trees or 150 12.5-cm-diameter trees provides 2 m^2 of residual basal area. Although the crowns of the larger-diameter trees are more expansive than those of the 12.5-cm trees, the latter would quickly respond to release. The shade cast by the combined crowns of these small-diameter trees could have a greater effect on the developing regeneration than would the shade cast by the larger trees.

The spatial distribution of residual trees within a harvest unit also can significantly affect regeneration stem densities. Residual basal area maintained in discrete patches will minimize shading of regenerating hardwoods and therefore effects of this shade on regeneration stem densities. Residual basal areas >4.9 m^2/ha can reduce regeneration stem densities and should not be maintained within harvest units designed to provide quality habitat for ruffed grouse (Thompson and Dessecker 1997). In addition, residual basal area levels <4.9 m^2/ha in aspen and other shade-intolerant forest types can reduce stem densities and habitat quality for ruffed grouse (Perala 1977).

Research in aspen forests managed on a 40-year rotation shows that small harvest units (1–2 ha) are more beneficial to ruffed grouse than larger harvest units (Gullion 1984*a*). The small harvest units are designed to provide ruffed grouse with patches of protective cover (6- to 15-year-old stands) interspersed with mature stands that provide male flower buds for grouse during winter. In other forest types managed using longer rotations (60–100 years), ruffed grouse can benefit from habitat interspersion, but they may not benefit from a pattern of small-block timber harvests to the same degree as in aspen forests. Scattered small-block harvest units on landscapes dominated by mature forest can provide patches of habitat for ruffed grouse, but these isolated islands likely provide only limited security from predators.

Woodcock

Habitat management beneficial to ruffed grouse generally benefits woodcock. However, because woodcock feed primarily on earthworms and other invertebrates, soil moisture and fertility, slope, aspect, and other site factors must be considered (Sepik et al. 1981, Roberts 1989). Habitat management in valleys and lower slopes is more beneficial to woodcock than management on dry upper and middle slopes (Liscinsky 1972). McAuley et al. (1996) recommended maintaining ≥25% of a unit in early successional habitat through clearcutting blocks >2 ha, or cutting 30-m-wide strips in mature forest on about a 40-year rotation. Stands of alders and similar moist-soil shrub species should be encouraged and maintained by cutting strips on a 20-year rotation across moisture gradients. Blueberry fields can be maintained through periodic burning, and fields and pastures can be maintained by mowing either completely or in strips.

On wintering areas, burning, mowing, herbicide application, tillage, and timber harvest can be used to create or maintain habitat for woodcock. Regeneration created by clearcutting small blocks provides excellent woodcock cover, although selective cutting also can be beneficial (Roberts et al.1984). Thinning pole stands can benefit woodcock by encouraging development of midstory and understory vegetation. Mature stands can be maintained in good woodcock cover by removing sufficient overstory canopy to allow light to reach the ground and promote dense shrub layers. Seed tree and shelterwood cuts also are beneficial silvicultural treatments.

Feeding is presumed to be a primary reason woodcock use roosting fields in winter. Pastures and fallow fields can be burned to remove dense vegetation, which will make them more attractive to woodcock for nocturnal roosting (Glasgow 1958). Mowing strips and patches in fall after growth has ceased can help create foraging areas for woodcock (Krementz and Jackson 1999). Brief periods of intense grazing by livestock can be used to set back plant succession and enhance use by woodcock, although intensively grazed fields that result in extensive areas of very short grass will receive little use (Krementz and Jackson 1999).

Pine plantations <3 m tall are used by woodcock, but once the canopy closes and shrub densities decline, they are not often used (Krementz and Pendleton 1994). Thinning closed-canopy stands so that light can reach the forest floor allows thickets of hardwood seedlings, blackberries (*Rubus* spp.), switch cane (*Arundinaria tecta)* and other plants to grow under the pines. Clearcuts provide nighttime habitat and also, within a few years, daytime cover. Shelterwood and seed-tree harvests create patches of suitable woodcock habitat, but they also provide roost sites for birds of prey (Krementz and Jackson 1999). Clearcutting small blocks in stands of bottomland hardwoods provides excellent woodcock cover (Roberts et al. 1984).

Likely future trends

The long-term declines in ruffed grouse and woodcock populations are likely the result of the degradation and loss of suitable early successional habitats (Owen et al. 1977, Dwyer et al. 1983, Straw et al. 1994, Thompson and Dessecker 1997). Habitat loss has been associated with urbanization, especially in the mid-Atlantic states, forest succession on the northern breeding areas, and drainage and land-use conversion on the wintering grounds (Straw et al. 1994).

To provide quality habitat for ruffed grouse and woodcock, timber harvest and other forest disturbance should remove sufficient basal area and stems from a stand to allow understory development. Timber harvest practices commonly used in the eastern United States leave residual basal areas that exceed the levels necessary to allow development of quality understory habitat. Guidelines to manage forested riparian areas that, *a priori*, preclude removal of substantial overstory vegetation may unnecessarily limit development of early successional habitat on these sites, which could provide important resources for ruffed grouse, woodcock, and other early successional wildlife. During the interval between the 2 most recent forest inventories (early 1970s to mid-1980s), more than 60% of the basal area was removed from only 4% of the forest stands in West Virginia and from 8% of the stands in New England (Gansner et al. 1990, Birch et al. 1992). On most sites, removal of <60% of the basal area is not adequate to establish quality habitat for ruffed grouse and woodcock.

Only 14% of the timberland in the eastern United States is in public ownership (Powell et al. 1993). Public land management agencies have responded to public concerns over proactive forest management by proposing significant reductions in levels of timber harvest and in the prescription of even-age regeneration methods (USFS 1995). Approximately 70% of the timberland in the eastern United States is in non-industrial private (NIPF) ownership. Birch (1996) reported that privately owned forest tracts <40 ha in size increased from 12.3 million ha (26.7% of private forest land) in 1978 to 22.9 million ha (43.6% of private forest land) in 1994. As the size of NIPF tracts decreases, so does the likelihood of timber harvest activity (Birch 1986, Roberts et al. 1986).

The forests of the East are maturing. New England forests currently are dominated by saw timber-sized trees, whereas the early successional seedling–sapling stands that woodcock and ruffed grouse require are becoming regionally scarce and as of 1988 composed only 8% of the timberland in the Northeast (Brooks and Birch 1988). This trend is consistent throughout much of the East (Trani et al. 2001). Declines in young forest are the result of changing management objectives and techniques, changing attitudes of landowners, a decline in farm abandonment, increased fire suppression, and increased urbanization (Brooks and Birch 1988, USFWS 1996). Societal attitudes toward timber harvest may continue to limit efforts to provide early successional habitats, thereby exacerbating the ongoing declines of ruffed grouse, American woodcock, and other wildlife dependent upon young forest habitats.

Literature cited

BACKS, S. E. 2000. Breeding indices of ruffed grouse—Spring 2000. Wildlife management and research notes. Indiana Department of Natural Resources, Bedford, USA.

BATEMAN, M. C. 1999. Status of woodcock in Canada. Canadian Wildlife Service, Sackville, New Brunswick, Canada.

BERG, W. 2000. Minnesota grouse and hares. Minnesota Department of Natural Resources, Grand Rapids, USA.

BIRCH, T. W. 1986. Forest-land owners of Maine, 1982. United States Department of Agriculture Forest Service Resource Bulletin NE-90, Broomall, Pennsylvania, USA.

BIRCH, T. W. 1996. Private forest-land owners of the northern United States, 1994. United States Department of Agriculture Forest Service Resource Bulletin NE-136, Radnor, Pennsylvania, USA.

BIRCH, T. W., D. A. GANSNER, S. L. ARNER AND R. H. WIDMANN. 1992. Cutting activity on West Virginia timberlands. Northern Journal of Applied Forestry 9: 146-148.

BOYD, R. A. 1990. Population density and habitat utilization of ruffed grouse in the southern Appalachians. Thesis, University of Tennessee, Knoxville, USA.

BOURGEOIS, A. 1977. Quantitative analysis of American woodcock nest and brood habitat. Woodcock Symposium 6: 109-118.

BRITT, T. L. 1971. Studies of woodcock on the Louisiana wintering ground. Thesis, Louisiana State University, Baton Rouge, USA.

BROOKS, R. T., AND T. W. BIRCH. 1988. Changes in New England forests and forest owners: implications for wildlife habitat resources and management. Transactions of the North American Wildlife and Natural Resources Conference 53: 78-87.

BRUGGINK, J. G., AND W. L. KENDALL. 1995. American woodcock harvest and breeding population status, 1995. United States Fish and Wildlife Service, Laurel, Maryland, USA.

BRYANT, J. P., AND P. J. KUROPAT. 1980. Selection of winter forage by subarctic browsing vertebrates: the role of plant chemistry. Annual Review of Ecological Systems 11: 261-285.

CARPENTER, E. M., M. H. HANSEN, AND D. M. ST. JOHN . 1986. The private forest landowners of Minnesota - 1982. United States Department of Agriculture Forest Service Resource Bulletin NC-95, St. Paul, Minnesota, USA.

CAUSEY, M. K. 1989. Alabama woodcock investigations. Final report, P-R Project W-44:6-10 Study II. Alabama Department of Conservation and Natural Resources, Montgomery, USA.

COON, R. A., B. K. WILLIAMS, J. S. LINDZEY, AND L. GEORGE. 1982. Examination of woodcock nest sites in central Pennsylvania. Pages 55-62 *in* T. J. Dwyer and G. L Storm, technical coordinators. Woodcock ecology and management. United States Fish and Wildlife Service, Wildlife Research Report 14, Washington, D.C., USA.

DESSECKER, D. R. 1997. Back to the future - Is the past a guide to a "healthy" forest landscape in the northern Great Lakes region? Transactions North American Wildlife and Natural Resources Conference 62: 469-478.

DHUEY, B. 2000. Wisconsin wildlife surveys, August 2000. Wisconsin Department of Natural Resources, Monona, USA.

DIMMICK, R. W., J. D. SOLE, W. G. MINSER, AND P. E. HALE. 1998. Response of ruffed grouse to forest management in the southern Appalachian Mountains. Southeast Association Fish and Wildlife Agencies 52: 294-302.

DUNFORD, R. D., AND R. B. OWEN, JR. 1973. Summer behavior of immature radio-equipped woodcock in central Maine. Journal of Wildlife Management 37: 462-468.

DWYER, T. J., D. G. MCAULEY, AND E. L. DERLETH. 1983. Woodcock singing-ground counts and habitat changes in the northeastern United States. Journal of Wildlife Management 47: 772-779.

DWYER, T. J., G. F. SEPIK, E. L. DERLETH, AND D. G. MCAULEY. 1988. Demographic characteristics of a Maine woodcock population and effects of habitat management. United States Fish and Wildlife Service, Fish and Wildlife Research 4, Washington, D.C., USA.

DYER, J. M., AND R. B. HAMILTON. 1977. Analysis of several site components of diurnal woodcock habitat in southern Louisiana. Woodcock Symposium 6: 51-62.

GANSNER, D. A., T. W. BIRCH, S. L. ARNER, AND S. J. ZARNOCH. 1990. Cutting disturbance on New England timberlands. Northern Journal of Applied Forestry 7: 118-120.

GLASGOW, L. L. 1958. Contributions to the knowledge of the ecology of the American woodcock (*Philohela minor* Gmelin) on the wintering range in Louisiana. Dissertation, Texas A&M University, College Station, USA.

GODFREY, G. A. 1975. Home range characteristics of ruffed grouse broods in Minnesota. Journal of Wildlife Management 39: 287-298.

GREGG, L. E., AND J. B. HALE. 1977. Woodcock nesting habitat in northern Wisconsin. Auk 94: 489-493.

GULLION, G. W. 1966. The use of drumming behavior in ruffed grouse population studies. Journal of Wildlife Management 30: 717-729.

GULLION, G. W. 1984*a*. Managing northern forests for wildlife. Ruffed Grouse Society, Coraopolis, Pennsylvania, USA.

GULLION, G. W. 1984*b*. Grouse of the north shore. Willow Creek, Oshkosh, Wisconsin, USA.

HAMEL, P. B., AND E. R. BUCKNER. 1998. How far could a squirrel travel in the treetops? A prehistory of the southern forest. Transactions North American Wildlife and Natural Resources Conference 63: 309-315.

HOLLIFIELD, B. K., AND R. W. DIMMICK. 1995. Arthropod abundance relative to forest management practices benefiting ruffed grouse in the southern Appalachians. Wildlife Society Bulletin 23: 756-764.

HUDGINS, J. E., G. L. STORM, AND J. S. WAKELEY. 1985. Local movements and diurnal-habitat selection by male American woodcock in Pennsylvania. Journal of Wildlife Management 49: 614-619.

JAKUBAS, W. J. AND G. W. GULLION. 1993. Use of quaking aspen flower buds by ruffed grouse: its relationship to grouse densities and bud chemical composition. The Condor 93: 473-485.

JOHNSGARD, P. A. 1973. Grouse and quail of North America. University of Nebraska, Lincoln, USA.

KEITH, L. B. 1963. Wildlife's ten-year cycle. University of Wisconsin, Madison, USA.

KEITH, L. B., AND D. H. RUSCH. 1989. Predation's role in the cyclic fluctuations of ruffed grouse. Acta Congress International Ornithology 19: 699-732.

KELLEY, J. R., JR. 2000. American woodcock population status, 2000. United States Fish and Wildlife Service, Laurel, Maryland, USA.

KEPPIE, D. M., AND R. M. WHITING, JR. 1994. American woodcock (*Scolopax minor*). 28 pages *in* A. Poole and F. Gill, editors. The Birds of North America, No. 100. The Academy of Natural Sciences, Philadelphia, Pennsylvania, and The American Ornithologists' Union, Washington, D.C., USA.

KINSLEY, K. R., AND G. L. STORM. 1988. Structural characteristics of woodcock nesting habitat. Journal of the Pennsylvania Academy of Science 62: 142-146.

KREMENTZ, D. G., AND G. W. PENDLETON. 1994. Diurnal habitat use of American woodcock wintering along the Atlantic coast. Canadian Journal of Zoology 72: 1945-1950.

KREMENTZ, D. G., AND J. J. JACKSON. 1999. Woodcock in the Southeast: natural history and management for landowners. University of Georgia, Cooperative Extension Service Bulletin 1183, Athens, USA.

KROHN, W. B., F. W. MARTIN, AND K. P. BURNHAM. 1974. Band recovery distribution and survival estimates of Maine woodcock. American Woodcock Workshop 5.

KROLL, J. C., AND R. M. WHITING. 1977. Discriminate function analysis of woodcock winter habitat in east Texas. Woodcock Symposium 6: 63-71.

KUBISIAK, J. F. 1978. Brood characteristics and summer habitats of ruffed grouse in central Wisconsin. Wisconsin Department of Natural Resources, Technical Bulletin No. 108, Madison, USA.

KUBISIAK, J. F. 1984. The impact of hunting on ruffed grouse populations in the Sandhill Wildlife Area. Pages 151-168 *in* W. L. Robinson, editor. Ruffed grouse management: state of the art in the early 1980's. The Wildlife Society, Bethesda, Maryland, USA.

KUBISIAK, J. F. 1985. Ruffed grouse habitat relationships in aspen and oak forests of central Wisconsin. Wisconsin Department of Natural Resources, Technical Bulletin 151, Madison, USA.

LISCINSKY, S. A. 1972. The Pennsylvania woodcock management study. The Pennsylvania Game Commission, Research Bulletin No. 171, Harrisburg, USA.

LITTLE, S. 1974. Effects of fire on temperate forests: Northeastern United States. Pages 225-250 *in* T. T. Kozlowski and C. E. Ahlgren, editors. Fire and ecosystems. Academic, New York, New York, USA.

MARTIN, F. W. 1964. Woodcock age and sex determination from wings. Journal of Wildlife Management 28: 287-293.

MARTIN, F. W., S. O. WILLIAMS, J. D. NEWSOM, AND L. L. GLASGOW. 1969. Analysis of records of Louisiana-banded woodcock. Annual Conference of Southeastern Association of Game and Fish Commissions 23: 85-96.

McAuley, D. G., J. R. Longcore, G. F. Sepik , and G. W. Pendleton. 1996. Habitat characteristics of American woodcock nest sites on a managed area in Maine. Journal of Wildlife Management 60:138–148.

Mendall, H. L., and C. M. Aldous. 1943. The ecology and management of the American woodcock. Maine Cooperative Wildlife Research Unit, University of Maine, Orono, USA.

Morgenweck, R. O. 1977. Diurnal high use areas of hatching-year female American woodcock. Woodcock Symposium 6:155–160.

Norman, G. W. 2000. 1999–2000 ruffed grouse population status in Virginia. Virginia Department of Game and Inland Fisheries, Wildlife Resource Bulletin No. 00-7, Richmond, USA.

Owen, R. B., Jr., J. M. Anderson, J. W. Artmann, E. R. Clark, T. G. Dilworth, L. E. Gregg, F. W. Martin, J. D. Newsom, and S. R. Pursglove. 1977. American woodcock (*Philohela minor* = *Scolopax minor* of Edwards 1974). Pages 149–186 *in* G. C. Sanderson, editor. Management of migratory shore and upland game birds in North America. International Association of Fish and Wildlife Agencies, Washington, D.C., USA.

Parris, R. W. 1986. Forest vegetation, earthworm (Lumbricidae), and woodcock (*Scolopax minor*) relationships. Dissertation, State University of New York, Syracuse, USA.

Perala, D. A. 1977. Aspen in the north central states. United States Department of Agriculture Forest Service, General Technical Report NC-36, St. Paul, Minnesota, USA.

Pettingill, O. S., Jr. 1936. The American woodcock *Philohela minor* (Gmelin). Memoirs Boston Society Natural History 9:169–391.

Powell, D. S., J. L. Faulkner, D. R. Darr, Z. Zhu and D. W. MacCleery. 1993. Forest resources of the United States, 1992. United States Department of Agriculture Forest Service, General Technical Report RM-234, Fort Collins, Colorado, USA.

Rabe, D. L. 1977. Structural analysis of woodcock diurnal habitat in northern Michigan. Woodcock Symposium 6:125–134.

Rabe , D. L., and H. H. Prince. 1982. Breeding woodcock use of manipulated forest-field complexes in the aspen community type. Pages 114–119 *in* T. J. Dwyer and G. L Storm, technical coordinators. Woodcock ecology and management. United States Fish and Wildlife Service, Wildlife Research Report 14, Washington, D.C., USA.

Robbins, C. T., T. A. Hanley, A. E. Hagerman, O. Hjeljord, D. L. Baker, C. C. Schwartz, and W. W. Mautz. 1987. Role of tannins in defending plants against ruminants: reduction in protein availability. Ecology 68:98–107.

Roberts, J. C., W. G. Tlusty, and H. C. Jordahl, Jr. 1986. The Wisconsin private non-industrial woodland owner: a profile. University of Wisconsin Cooperative Extension Service, Madison, USA.

Roberts, T. H. 1989. American woodcock (*Scolopax minor*). Department of Defense Natural Resources Program, Technical Report EL-89-5. Department of the Army, Waterways Experiment Station, Corp. of Engineers, Vicksburg, Mississippi, USA.

Roberts, T. H. 1993. The ecology and management of wintering woodcocks. American Woodcock Symposium 8:87–97.

Roberts, T. H., E. P. Hill, and E. A. Gluesing. 1984. Woodcock utilization of bottomland hardwoods in the Mississippi Delta. Annual Conference of the Southeastern Association of Fish and Wildlife Agencies 38:137–141.

Robinson, S. K., F. R. Thompson, III, T. M. Donovan, D. R. Whitehead, and J. Faaborg. 1995. Regional forest fragmentation and the nesting success of migratory birds. Science 267:1987–1990.

Sauer, J. R., and J. B. Bortner. 1991. Population trends from the American woodcock singing ground survey, 1970–88. Journal of Wildlife Management 55:300–312.

Sepik, G. F., R. B. Owen, Jr., and M. W. Coulter. 1981. A landowner's guide to woodcock management in the northeast. Maine Agricultural Experiment Station, Miscellaneous Report 253, Orono, USA.

Sepik, G. F. and T. J. Dwyer. 1982. Woodcock response to habitat management in Maine. Pages 106–113 *in* T. J. Dwyer and G. L. Storm, editors. Woodcock ecology and management. United States Fish and Wildlife Service, Wildlife Research Report 14, Washington, D.C., USA.

Sepik, G. F., R. B. Owen, Jr., and T. J. Dwyer. 1983. The effect of drought on a local woodcock population. Transactions of Northeast Fish and Wildlife Conference 40:1–8.

Sepik, G. F., and E. L. Derleth. 1993. Habitat use, home range size and patterns of moves of the American woodcock in Maine. American Woodcock Symposium 8:41–49.

Sepik, G. F., D. G. McAuley, and J. R. Longcore. 1993. Critical review of the current knowledge of the biology of the American woodcock and its management on the breeding grounds. American Woodcock Symposium 8:98–104.

Servello, F. A., and R. L. Kirkpatrick. 1987. Regional variation in the nutritional ecology of ruffed grouse. Journal of Wildlife Management 51:749–770.

Sheldon, W. G. 1967. The book of the American woodcock. University of Massachusetts, Amherst, USA.

Smith, H. C., N. I. Lamson, and G. W. Miller. 1989. An esthetic alternative to clearcutting? Deferment cutting in eastern hardwoods. Journal of Forestry 87:14–18.

Sperry, C. C. 1940. Food habits of a group of shorebirds: woodcock, snipe, knot, and dowitcher. United States Biological Survey, Wildlife Research Bulletin 1, Washington, D.C., USA.

Stoll, R. J., Jr., M. W. McClain, R. L. Boston, and G. P. Honchul. 1979. Ruffed grouse drumming site characteristics in Ohio. Journal of Wildlife Management 43:324–333.

Stoll, R. J., Jr., W. L. Culbertson, M. W. McClain, R. W. Donohue, and G. Honchul. 1999. Effects of clearcutting on ruffed grouse in Ohio's oak-hickory forests. Ohio Department of Natural Resources, Fish and Wildlife Report 14, Columbus, USA.

Straw, J. A., Jr., D. G. Krementz, M. W. Olinde, and G. F. Sepik. 1994. American woodcock. Pages 97–114 *in* T. C. Tacha and C. E. Braun, editors. Migratory shore and upland game bird management in North America. International Association of Fish and Wildlife Agencies, Washington, D.C., USA.

Thompson, F. R., III, D. A. Freiling, and E. K. Fritzell. 1987. Drumming, nesting, and brood habitats of ruffed grouse in an oak-hickory forest. Journal of Wildlife Management 51:568–575.

Thompson, F. R., III, and E. K. Fritzell. 1988. Ruffed grouse winter roost site preference and influence on energy demands. Journal of Wildlife Management 52:454–460.

Thompson, F. R., III, and D. R. Dessecker. 1997. Management of early-successional communities in central hardwood forests. United States Department of Agriculture Forest Service General, Technical Report NC-195, St. Paul, Minnesota, USA.

Trani, M. K., R. T. Brooks, T. L. Schmidt, V. A. Rudis, and C. M. Gabbard. 2001. Patterns and trends of early successional forests in the eastern United States. Wildlife Society Bulletin 29:413–424.

United States Department of the Interior. 1990. American woodcock management plan. United States Fish and Wildlife Service, Washington, D.C., USA.

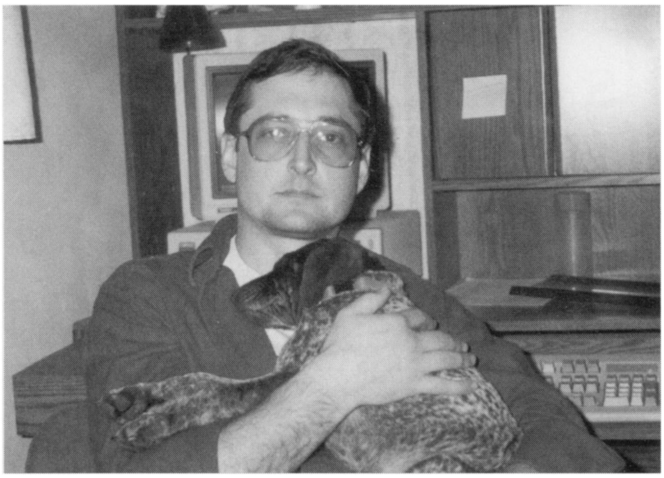
United States Department of the Interior. 1997. 1996 National survey of fishing, hunting, and wildlife-associated recreation. United States Department of Interior, Washington, D.C., USA.

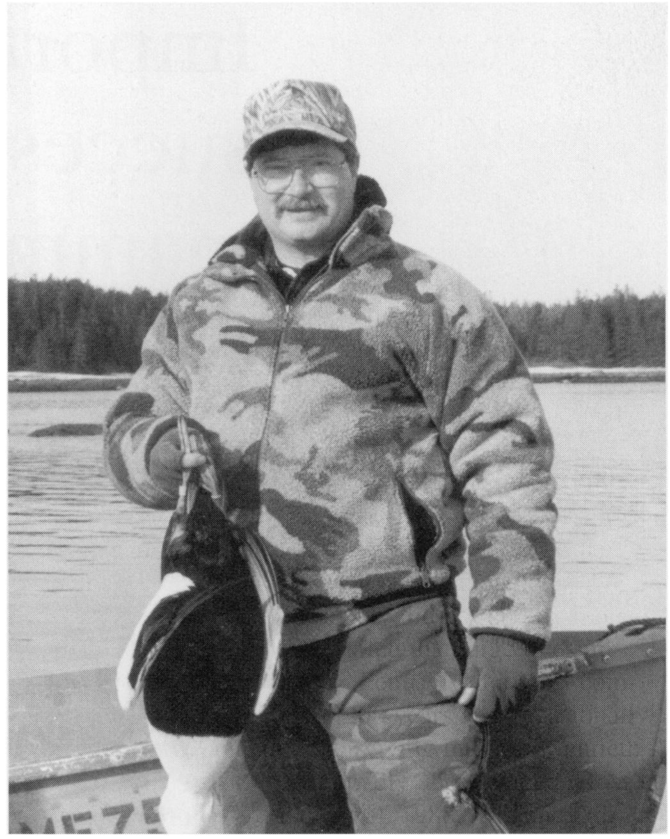
United States Fish and Wildlife Service. 1996. American woodcock management plan. United States Fish and Wildlife Service, Region 5, Hadley, Massachusetts, USA.

United States Forest Service. 1995. The Forest Service program for forest and rangeland resources: a long-term strategic plan. United States Department of Agriculture Forest Service, Washington, D.C., USA.

Weigel, D. R. and G. R. Parker. 1995. Tree regeneration following group selection harvesting in southern Indiana. K. W. Gottschalk and S. L. Fosbroke, editors. Central Hardwood Forest Conference 10: 316–325.

WHITCOMB, S. D., V. R. TUOVILA, AND C. A. STEWART. 2000. Ruffed grouse and American woodcock status in Michigan, 2000. Michigan Department of Natural Resources, Report No. 3318, Lansing, USA.





Dan Dessecker (left) is the senior wildlife biologist for the Ruffed Grouse Society. Dan received a B.S. (1981) from the University of Wisconsin–Stevens Point and an M.S. (1984) from Pennsylvania State University. His professional interests include the historical and current roles of disturbance in forest ecology. Dan is past-president of the Wisconsin Chapter of The Wildlife Society. ***Dan McAuley*** (right) is a research wildlife biologist with the USGS Patuxent Wildlife Research Center, stationed in Orono, Maine, where he has worked since 1978. He received a B.A. from Niagara University and a B.S. and M.S. in wildlife management from the University of Maine, Orono. He has been involved in research on breeding ecology and management of American woodcock during his entire career. His other research interests include brood ecology of waterfowl in the Northeast; survival, habitat use, and movements of American black ducks; interspecific aggression and brood productivity of sympatric black ducks and mallards; and use of managed habitats by songbirds. He is currently working on a multi-state study of woodcock survival in the northeast U.S.

Associate editors: *DeGraaf, Thompson, and Trani*